

Medical documents Translation Service

User & Developer Overview

What is medii.ai?

medii.ai is a medical document translation platform designed to translate Hebrew medical documents into English while preserving the original document structure, formatting, and layout.

The system combines AI-powered translation with automated quality assurance and DOCX document generation.

Core Workflow

1. Create a Project

Users create a translation project by providing:

- Project name
- Deadline
- Priority level

Each project receives a unique identifier and workspace.

2. Upload Documents

Required

- Hebrew source DOCX document

Optional (Recommended)

- English reference DOCX document

The system automatically extracts document metadata such as page count and word count.

3. Parse Documents

The uploaded source document is analyzed and divided into structured sections.

The parser extracts:

- Headings
- Paragraphs
- Lists
- Tables
- Formatting information
- Images and layout structure

Reference documents are indexed to improve terminology consistency.

4. AI Translation

The platform uses OpenAI GPT-4 to translate content section by section.

Key features:

- Medical terminology preservation
 - Reference-aware translation
 - Consistent wording across the document
 - Real-time progress tracking
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5. Quality Assurance

After translation, automated QA checks are performed.

Checks include:

- Terminology consistency
- Structural accuracy
- Reference coverage
- Medical content validation

The system generates a QA report highlighting any issues.

6. Document Generation

A final English DOCX file is produced automatically.

The generated document:

- Preserves formatting
- Mirrors the source structure

- Maintains tables and lists
 - Uses translated content from approved sections
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Technology Stack

Frontend

- Next.js 14
- React 18
- TypeScript
- Tailwind CSS
- Radix UI

Backend

- FastAPI
- Python
- Uvicorn

AI & Processing

- OpenAI GPT-4
 - python-docx
 - BeautifulSoup4
 - lxml
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Main Features

- ✓ Hebrew → English medical translation
- ✓ DOCX upload and export
- ✓ Formatting preservation
- ✓ Reference document alignment
- ✓ Automated quality assurance
- ✓ Project management dashboard
- ✓ User authentication
- ✓ Translation review editor

Project Structure

The application consists of:

Frontend

Handles:

- Authentication
- Project management
- Uploads
- Translation workflow
- QA review

Backend

Handles:

- Document parsing
- Translation jobs
- QA generation
- DOCX rendering
- File management

Environment Requirements

Node.js

Version 18+

Python

Version 3.8+

Required Environment Variables

Backend: OPENAI_API_KEY

Frontend: NEXT_PUBLIC_API_URL

Current Status

The platform is production-ready and includes:

- Complete translation workflow
- Real-time job tracking
- DOCX parsing and rendering
- QA reporting
- Authentication system

Additional features such as collaboration tools, role management, and translation memory may be added in future releases.
